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Displacement prediction strategy based on KCF in tracking

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Abstract

Real-time and high precision tracking of moving targets is always a challenging subject in the field of target tracking. In this paper, based on the feature extraction and detection method of kernel correlation filtering algorithm KCF, target displacement prediction is introduced into the algorithm to achieve more accurate tracking effect. In addition, the scale pool and multi-feature (descriptor) strategy of SAMF algorithm are introduced and improved to achieve real-time tracking of moving targets with higher precision. The experimental results show that the proposed algorithm achieves a good tracking effect on moving targets, and has a good performance in tracking accuracy and tracking speed.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision; Computer vision problems; Tracking.

Keywords

KCF, SAMF, Displacement prediction, multi-feature, multi-scale

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1 INTRODUCTION

In recent years, with the rapid development of computer technology, the field of computer vision has made great progress, including the field of target tracking. The use of visual target tracking technology to achieve real-time tracking of moving targets or continuous tracking of specific targets in video files has a wide range of application scenarios and application value. It is becoming more and more normal to use target tracking technology on various unmanned platforms, and the application of target tracking algorithm to ground unmanned vehicle platform has quite high application value.

There are two kinds of visual object tracking methods: generative class tracking method and discriminant class tracking method. Since the generated class tracking method can not adapt to the complex and changeable tracking scene, it is gradually reduced in use.

In the discriminant class tracking method, algorithms based on correlation filtering and deep learning[1][2][3] have been widely used. Correlation filter tracking algorithm is a kind of tracking algorithm which can learn and update the model in real time, and can realize real-time tracking[4][5]. MOSSE[6] algorithm introduces correlation filtering algorithm into the field of target tracking, MOSSE algorithm goal is to train a filter, so that the square error between the cyclic correlation output and the desired label is minimal, and the Fourier transform is used to transform the problem into the frequency domain, and get a high-speed tracking effect, which can reach more than 600 frames per second. Then on the basis of the MOSSE also developed many new algorithms, such as the CN[7], CSK[8], KCF[9], DSST[10], SAMF[11], AutoTrack[12][13], etc. CT[14] algorithm is a fast, robust and simple method in discriminant class tracking. CT algorithm uses the theory of compressed sensing to classify and discriminate features after dimensionality reduction to achieve tracking effect. CN algorithm uses color name feature to describe target feature, which improves the effect of tracking algorithm. CSK algorithm uses ridge regression, cyclic matrix, fast Fourier transform and other methods to accelerate the calculation and training to get the filter, CSK algorithm uses gray features. Compared with CSK algorithm, KCF algorithm introduces HOG feature, proposes multi-channel sample training strategy, and carries on some optimization to accelerate the calculation. Both DSST and SAMF algorithms put forward their own solutions to the problem of scale change of tracking targets, and successfully improve the accuracy and success rate of tracking, but the speed of tracking algorithms has decreased. AutoTrack algorithm proposes an online automatic and adaptive learning method of relevant parameters, which can effectively use the local and global information hidden in the response graph. In discriminant class tracking methods, many algorithms using deep learning methods have emerged in recent years, but these methods are usually difficult to meet the requirements of real-time tracking. At the same time, due to the hardware limitations of on-board calculators on mobile devices, it is difficult to run these deep learning-based target tracking algorithms well.

Now target tracking algorithm has a good performance in many scenes[15][16], but there are still some problems, such as tracking target scale change, target deformation and occlusion problems. These problems may often be encountered when running object tracking algorithms on unmanned platforms such as unmanned vehicles[17]. In this case it is important to find an algorithm that can solve these problems to a certain extent and does not require too much computing power. The performance of unmanned platform, such as the on-board computer of unmanned vehicle, is relatively limited. In this case, it is a meaningful work to find a tracking

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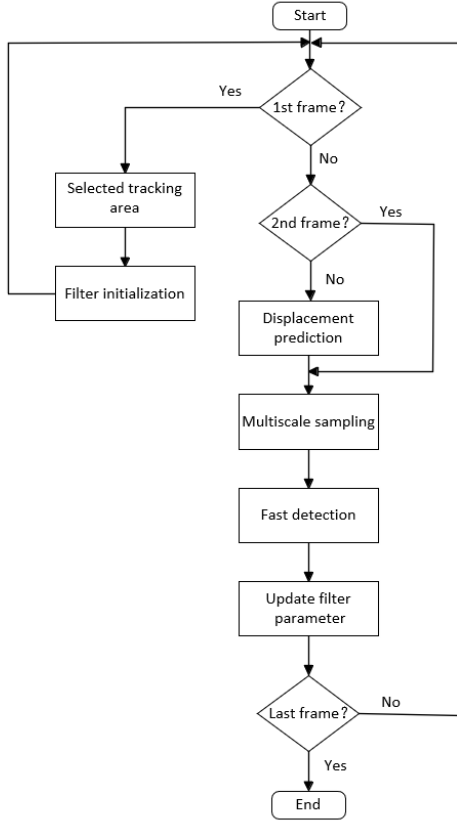


Figure 1: Flow chart of tracking algorithm.

algorithm that can ensure the tracking effect and achieve the real-time tracking speed.

Based on KCF algorithm, this paper obtains an improved algorithm, which retains HOG feature, introduces gray feature and color name feature, introduces multi-scale pool strategy, and introduces target displacement prediction. This ensures the tracking speed of the target tracking algorithm and improves the tracking accuracy and success rate compared with the KCF algorithm, which is a good real-time tracking target tracking algorithm.

2 TRACKING ALGORITHMS

This paper presents an improved KCF algorithm which introduces displacement prediction function, multi-feature and scale pool strategy. Select the tracking target region in the first frame, calculate the characteristic value and relevant parameters according to the target region, predict the target operation according to the target displacement difference between the present frame and the previous frame from the second frame, adjust the target detection region, and calculate the response value of the target region of different sizes according to the scale pool, select the region with the largest response value, no matter the first frame or the next frame. The features used are the integration of HOG feature, gray feature and CN feature. The algorithm flow chart is shown in Figure 1.

The following introduces the principles of KCF algorithm and SAMF algorithm, which is the basis of the improved algorithm proposed in this paper.

2.1 Principle of KCF algorithm

KCF algorithm is a target tracking algorithm based on and correlation filters proposed in 2014. It has a wide range of applications in target tracking, with high efficiency, accuracy and robustness. KCF algorithm is based on discriminant tracking framework, uses kernel function to build correlation filter, and tracks the target through training samples. KCF algorithm generates multiple copies through cyclic matrix, and uses each sample for regression training, thus enhancing the performance of the tracker. KCF algorithm mainly includes these parts: cyclic matrix, ridge regression, kernel technique, feature extraction, fast detection and model updating.

2.1.1 Cyclic Matrix. Let's say we have a vector represented by x , $x = [x_1, x_2, x_3, \dots, x_n]^T$, and this vector represents information about the object of interest, we call this vector base sample. Our goal is to train a classifier using this basic sample and the many samples that are used as negative samples using this basic sample transformation. P is the permutation matrix used to generate negative samples.

$$P = \begin{bmatrix} 0 & 0 & 0 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix} \quad (1)$$

Take the permutation matrix and x product, shift $Px = [x_n, x_1, x_2, \dots, x_{n-1}]^T$, by the matrix power $P^u x$ can get a larger shift.

$$\{P^u x | u = 0, \dots, n-1\} \quad (2)$$

In this way, the circulation-based matrix can be obtained.

$$X = \begin{bmatrix} x_1 & x_2 & x_3 & \cdots & x_n \\ x_n & x_1 & x_2 & \cdots & x_{n-1} \\ x_{n-1} & x_n & x_1 & \cdots & x_{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_2 & x_3 & x_4 & \cdots & x_1 \end{bmatrix} \quad (3)$$

The cyclic matrix can be made diagonal by discrete Fourier transform (DFT).

$$X = F \text{diag}(\hat{x}) F^H \quad (4)$$

where F is the constant matrix, $\hat{x}$ is the vector obtained by DFT, and $\hat{x} = \mathcal{F}(x)$. The cyclic matrix can be diagonalized in Fourier space. KCF algorithm uses this property to reduce the computation and improve the running speed of the algorithm.

2.1.2 Ridge Regression. Ridge regression is an improved least square method, which introduces regularization terms based on least square method. When using a linear model, $f(z) = \omega^T z$, this function can minimize the square error of sample x_i and regression target y_i .

$$\min_{\omega} \sum_i (f(x_i) - y_i)^2 + \lambda \|\omega\|^2 \quad (5)$$

Taking the partial derivative of ω yields:

$$\omega = (X^T X + \lambda I)^{-1} X^T y \quad (6)$$

Since all subsequent calculations are performed in the Fourier domain, where values are generally complex numbers, the formula becomes:

$$\omega = (X^H X + \lambda I)^{-1} X^H y \quad (7)$$

Since X is a cyclic matrix, we can further obtain:

$$\hat{\omega} = \frac{\hat{x}^* \odot \hat{y}}{\hat{x}^* \odot \hat{x} + \lambda} \quad (8)$$

2.1.3 Kernel Trick. When dealing with data, nonlinear data is often encountered. KCF algorithm uses kernel trick to map linearly indivisible data into high-dimensional space and make it linearly divisible. If the mapping function is $\varphi(x)$, then:

$$\omega = \sum_i \alpha_i \varphi(x_i) \quad (9)$$

The solution to equation (5) becomes:

$$\min_{\alpha} \left\| \varphi(X) \varphi(X)^T \alpha - y \right\|^2 + \lambda \left\| \varphi(X)^T \alpha \right\|^2 \quad (10)$$

Taking the derivative of α and setting the derivative equal to 0, we get:

$$\begin{aligned} \alpha &= (K + \lambda I)^{-1} y \\ &= \left(\varphi(X) \varphi(X)^T + \lambda I \right)^{-1} y \end{aligned} \quad (11)$$

where K is the kernel matrix, KCF algorithm generally uses Gaussian kernel function to calculate, Gaussian kernel function makes the kernel matrix is a cyclic matrix.

$$\begin{aligned} k(x, x') &= \exp \left(-\frac{1}{\sigma^2} \|x - x'\|^2 \right) \\ &= \exp \left(-\frac{1}{\sigma^2} \left(\|x\|^2 + \|x'\|^2 - 2\mathcal{F}^{-1}(\hat{x}^* \odot \hat{x}') \right) \right) \end{aligned} \quad (12)$$

Eventually, we can get it.

$$\hat{\alpha} = \frac{\hat{y}}{(\hat{k}^{xx} + \lambda)} \quad (13)$$

2.1.4 Feature Extraction. KCF algorithm extracts and characterizes the target information by using FHOG features reduced by PCA, and finally obtains 31-dimensional features. The resulting FHOG features provide a method to integrate high-dimensional features into the correlation filter framework through pixel-level feature mapping and discretization of gradient direction, and have a good performance in target tracking tasks.

2.1.5 Fast Detection. KCF algorithm also simplifies and accelerates the detection process, using K^z to represent the kernel matrix between all training samples and candidate regions, then:

$$K^z = C(k^{xz}) \quad (14)$$

From formula (9), it can be obtained:

$$f(z) = (K^z)^T \alpha \quad (15)$$

Fourier transform diagonalization gives:

$$\hat{f}(z) = \hat{k}^{xz} \odot \alpha \quad (16)$$

2.1.6 Model Update. KCF algorithm calculates the current model $\hat{\alpha}$ in the first frame, and records the current model α' in each subsequent frame. The model of each subsequent frame is obtained by linear interpolation between the current model and the previous model, with a learning rate of η and a size between 0 and 1. Then:

$$\alpha_{new} = (1 - \eta) \alpha_{old} + \eta \alpha' \quad (17)$$

2.2 Principle of SAMF algorithm

Based on KCF algorithm, since kernel correlation function only needs to compute dot product and vector norm, multi-channel image features can be used. Assuming that multiple channels of feature data are connected into a vector $x = [x_1, x_2, \dots, x_C]$, then equation (12) can be changed to:

$$k(x, x') = \exp \left(-\frac{1}{\sigma^2} \left(\|x\|^2 + \|x'\|^2 - 2\mathcal{F}^{-1}(\sum_C \hat{x}_c^* \odot \hat{x}'_c) \right) \right) \quad (18)$$

This allows the algorithm to use the fusion of multiple features to get better results. In addition to the gray pixels of the original image, the SAMF algorithm also uses HOG features and CN features. HOG feature and CN feature complement each other. HOG feature focuses on gradient change of image, while CN feature focuses on color information.

In addition, for the case that KCF can't handle the scaling change in video, SAMF algorithm adds a scale adaptive method to deal with the scaling change of tracking target. Suppose that the sample size is $S_T = (s_x, s_y)$, the scale pool is $S = \{t_1, t_2, \dots, t_k\}$, and the original target window size is S_t , then in the current frame, SAMF algorithm samples k sizes in $\{t_i S_t | t_i \in S\}$ to find the appropriate size, and after adjusting the size to S_T , the final response value is calculated by the following formula:

$$\arg \max F^{-1} \hat{f}(z^{t_i}) \quad (19)$$

where z^{t_i} is the sample size of $t_i S_T$, adjusted to S_T . After obtaining a vector by the corresponding function, by calculating its largest scalar, find the subscript of the maximum value, and get t_i . The scaling of tracking objects in video is handled by adaptive change and selection of scale pool.

2.3 Improved Algorithm

2.3.1 Displacement Prediction. The object tracked by the unmanned platform often moves at a constant speed or changes quickly in a short period of time. In this case, the displacement between the current frame and the last frame can be estimated according to the displacement changes of the first two frames. Assuming uniform linear motion, the position of the current frame is as follows:

$$P_t = P_{t-1} + (P_{t-1} - P_{t-2}) \quad (20)$$

where P_t is the position of the current frame, P_{t-1} is the position of the previous frame, and P_{t-2} is the position of the target two frames ago.

In the KCF algorithm, the current target region is multiplied by coefficient padding to get the region to be detected. If the target moves too fast, it may move out of the region to be detected, and it is more likely to move to the edge of the region to be detected. However, the region to be detected is itself used by superimposed

cosine window. When the target moves too fast to the edge of the area to be detected, the target information is likely to be lost, resulting in tracking failure.

In practical application, when the displacement of the first two frames is too small, it is considered not to perform position estimation, and two thresholds μ_1 and μ_2 are set. At the same time, considering that the tracking target is not moving uniformly in a straight line, the displacement of the first two frames is multiplied by a coefficient κ and then the position estimation can cope with a variety of motion situations of the object. Therefore, the position estimation of the current frame based on displacement prediction is:

$$P_t = P_{t-1} + \kappa (P_{t-1} - P_{t-2}), \mu_1 < |P_{t-1} - P_{t-2}| < \mu_2 \quad (21)$$

2.3.2 Scale Estimation. Drawing on and learning the idea of SAMF algorithm's scale pool, this paper sets up a 3-scale scale pool to carry out traversal calculation and comparison of tracking targets to find the most appropriate scale size for each frame. Because SAMF algorithm sets a 7-scale scale pool, it will take more time to traverse calculation and comparison. The 3-scale scale pool in this paper improves the speed of the tracking algorithm without affecting the accuracy and success rate of tracking too much.

Assuming that the sample size is $S_T = (s_x, s_y)$, the scale pool is $S = \{t_1, t_2, \dots, t_k\}$, and the interval between different scales of the SAMF algorithm is s_δ , then the scale pool S setting of the SAMF algorithm is:

$$S = \{1 - 3s_\delta, 1 - 2s_\delta, 1 - s_\delta, 1, 1 + s_\delta, 1 + 2s_\delta, 1 + 3s_\delta\} \quad (22)$$

The scale pool Settings in this paper are:

$$S_{\text{pool}} = \{1 - 2s_\delta, 1, 1 + 2s_\delta\} \quad (23)$$

After such adjustment, the algorithm can make effective adjustment to the scale change of the tracking target without wasting too much computing resources and time.

2.3.3 Multi-Feature. To learn the multi-feature fusion of SAMF algorithm, this paper uses FHOG feature, gray feature and CN feature to improve the tracking effect of tracking targets through the fusion of multi-feature information. FHOG feature extracts the gradient information of tracking targets, gray feature and CN feature show the color features of tracking targets, and FHOG has 31-dimensional information. Gray feature and CN feature have a total of 11 dimensions of information, and the fused feature has a total of 42 dimensions, which does not increase much information compared with the original FHOG feature information, but plays a good role.

3 EXPERIMENTS

All tracking algorithms were run on Windows 10, 64bit systems using MATLAB R2022b simulation software. The CPU of this computer is Intel i9 13900KF, the running frequency is 3GHz, the GPU is NVIDIA GeForce RTX 4090, the video memory is 24GB, the memory is 32GB.

The data set tested by the algorithm in this paper is OTB100, which is a data set established in 2015. It has 98 video sequences and 100 tracking targets, including both grayscale images and color images. OTB100 has 11 different attribute categories, They are illumination variation (IV), scale variation (SV), Occlusion (OCC),

deformation (DEF), motion blur (MB), fast motion (FM), in-plane rotation (IPR), out-of-plane rotation (OPR), out-of-field of view (OV), background clutter (BC), and low resolution (LR).

3.1 Evaluation Metrics and Methodology for Algorithms

OTB data sets are evaluated in One Pass Evaluation (OPE), Temporally Robust Evaluation (TRE) and Spatial Robust Evaluation (SRE). In this paper, OPE evaluation method is adopted. OPE method initializes the first frame with ground-truth target position, and then runs tracking algorithm to get average accuracy and success rate.

Accuracy refers to the accuracy of tracking the target's position. In the evaluation of tracking accuracy, the center position error is a widely used criterion, which is the average pixel distance (Euclidean distance) between the center position of the tracking target and the exact position manually labeled. In general, the average center position error of all frames in a sequence is used to evaluate the overall performance of the tracking algorithm for the sequence. However, when the tracker loses the target, the predicted tracking position is random, and the average error value may not accurately evaluate the performance of the tracker. Therefore, the OTB data set was further expanded into an accuracy curve to calculate the successful tracking ratio under different threshold distances, and the value corresponding to a threshold of 20 pixels was adopted as a representative accuracy evaluation index.

Success rate refers to the percentage of success in the tracking area. Because the center position error cannot evaluate the scale change of the target in the tracking process, an evaluation criterion based on the area overlap ratio is proposed, and the performance of the tracking algorithm is measured by the area overlap ratio between the bounding box obtained by the statistical tracking algorithm and the accurate bounding box marked manually. Firstly, overlap score (OS) is defined. bounding box (denoted as a) obtained by tracking algorithm and box (denoted as b) given by ground-truth are defined as:

$$OS = \left| \frac{a \cap b}{a \cup b} \right| \quad (24)$$

When the OS of a certain frame is greater than the set threshold, the frame is considered a Success, and the percentage of the total successful frames in all frames is the Success rate. The value of OS ranges from 0 to 1, so you can plot the success rate of the threshold change from 0 to 1. Then, the Area Under Curve (AUC) of the success rate curve is used to measure the success rate of the tracking algorithm.

3.2 Qualitative Analysis

In this part, Human6 sequence, boy sequence and Car1 sequence in OTB data set are taken as examples to compare and analyze the effect of the tracking algorithm. Human6 sequence has 5 attributes, such as scale transformation, occlusion, deformation, fast movement and out of view. boy sequence has 5 attributes: scale change, motion blur, fast motion, in-plane rotation, out-of-plane rotation, Car1 has 4 attributes: illumination change, scale change, background clutter, low resolution. There are 6 tracking algorithms for comparison, which are the present algorithm, CT algorithm,



Figure 2: Tracking effects of different sequences in OTB data set.

CSK algorithm, KCF algorithm, DSST algorithm and SAMF algorithm. The tracking area of these algorithms is represented by a square box, among which the algorithm in this paper is represented by a red box, the CT algorithm is represented by a green box, the CSK algorithm is represented by a blue box, the KCF algorithm is represented by a black box, the DSST algorithm is represented by a pink box, and the SAMF algorithm is represented by a cyan box. In Figure 2, the first row represents the tracing effect of the Human6 sequence, the second row represents the tracing effect of the boy sequence, and the third row represents the tracing effect of the Car1 sequence.

At the beginning of the Human6 sequence, CT algorithm did not track the target, then CSK algorithm and KCF algorithm also lost track of the target, and then around 394 frames, DSST algorithm also lost track of the target, and finally SAMF algorithm also lost track of the target, only the algorithm in this paper is still tracking. This shows that the algorithm in this paper has excellent performance in 5 attributes such as fast motion, scale change and deformation. In boy sequence and Car1 sequence, the proposed algorithm, DSST algorithm and SAMF algorithm can track the target well, which shows that the proposed algorithm also has good performance in motion ambiguity, in-plane rotation, low resolution and other attributes. Based on the above analysis, the algorithm in this paper has a good performance in processing the fast motion and scale transformation of the tracking target, and also has a good effect on other attributes.

3.3 Quantitative Analysis

In this part, six algorithms including the proposed algorithm, CT algorithm, CSK algorithm, KCF algorithm, DSST algorithm and

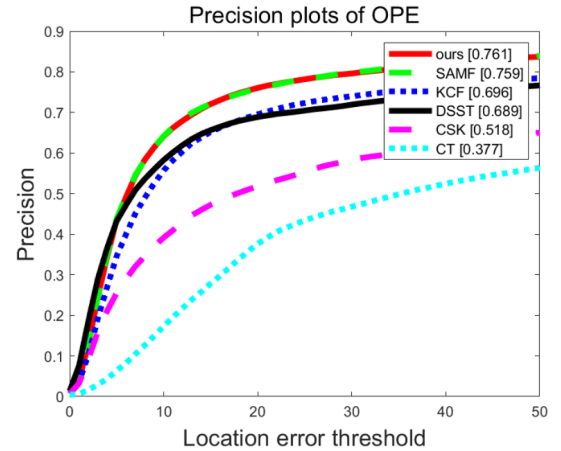


Figure 3: Precision.

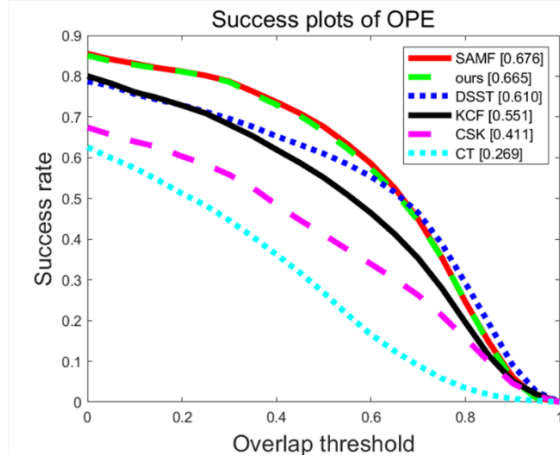
SAMF algorithm are tested on the whole OTB data set, and the results are compared. The experimental results are shown in Figure 3, Figure 4 and Table 1.

Figure 3 shows the accuracy of these 6 algorithms tested on OTB100 data set. The results show that compared with KCF algorithm, the effectiveness of the proposed algorithm is improved by 6.5%, and compared with SAMF algorithm, the proposed algorithm is improved by 0.2%.

Figure 4 shows the success rate of these 6 algorithms tested on OTB100 data set. Compared with KCF algorithm, the effectiveness of this algorithm increased by 11.4%, while compared with SAMF

Table 1: Speed of the algorithms.

Algorithms	CT	CSK	KCF	DSST	SAMF	Ours
Speed(frame/s)	324	968	404	59	48	92

**Figure 4: Success rate.**

algorithm, the difference between this algorithm and SAMF algorithm is 1.1%. In combination with Table 1, it can be seen that the algorithm in this paper runs nearly twice as fast as SAMF algorithm, which indicates that the algorithm in this paper has better real-time performance when the accuracy rate is slightly improved compared with SAMF algorithm and the success rate is not different.

4 CONCLUSION

In this paper, an improved target tracking algorithm based on KCF algorithm is proposed. The scale pool module and multi-feature method of SAMF algorithm are learned by reference, and the displacement prediction module is added. The experimental results show that the proposed algorithm has excellent performance compared with CSK algorithm, KCF algorithm, DSST algorithm, SAMF algorithm, etc., and can track the target well when the target is fast moving and the scale changes. Compared with KCF algorithm, the proposed algorithm can improve the performance and ensure the running speed of the algorithm.

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